GS-LSS: Learned Stochastic Sampling for 3D Gaussian Splatting under Biased Viewpoints

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***Abstract*—3D Gaussian Splatting (3DGS) achieves high-fidelity rendering when trained with sufficient observations and complete scene coverage. However, real-world data acquisition is often subject to *biased viewpoints*, including viewing angle disparities, discontinuous trajectories, and systematic coverage gaps. Under such bias, per-frame 3DGS optimization becomes underdetermined in weakly observed regions, where Gaussian primitives can collapse onto globally inconsistent local minima, leading to *perspective overfitting* that manifests as ghosting artifacts, hollow structures, and floating noise in unseen views. We argue that addressing this challenge requires regularizing the per-frame objective with an observed-support consistency criterion, rather than attempting to recover unobservable geometry solely from biased observations. To this end, we propose GS-LSS, an observed-support regularized learned stochastic optimization framework that reformulates 3DGS training as a Learned Stochastic Sampling (LSS) process, combining a deterministic drift that regularizes Gaussian primitives toward aggregated observed support with adaptive stochasticity that explores weakly constrained regions. The observed support is instantiated by a Continuous Probability Field (CPF), a continuous, globally aggregated structural field that serves as a data-adaptive consistency prior for Gaussian relocation and reshaping. Importantly, CPF is not designed to hallucinate fully unseen geometry. Instead, it estimates reliable observed support as a stabilizer whose approximation error acts as a local perturbation to the regularized objective, thereby reducing local sampling bias and suppressing view-specific overfitting artifacts. Experiments on three representative biased viewpoint scenarios show that GS-LSS achieves competitive or superior quantitative results among the evaluated Gaussian Splatting baselines, particularly under severe sampling bias.**

***Index Terms*—Biased Viewpoints, Gaussian Splatting, Learned Stochastic Sampling, Continuous Probability Field, Observed Support Regularization**

I. Introduction

NOVEL view synthesis has achieved significant progress since the introduction of 3D Gaussian Splatting (3DGS) [1]. The discrete representation and pixel-wise optimization scheme enable high-fidelity real-time rasterization at reasonable cost. Consequently, numerous variants have been proposed to integrate 3DGS into real-world applications, such as virtual tourism and cultural heritage preservation. The quality of scene reconstruction is strongly correlated with the completeness of data sampling, where broader viewpoint coverage typically leads to more faithful rendering quality.

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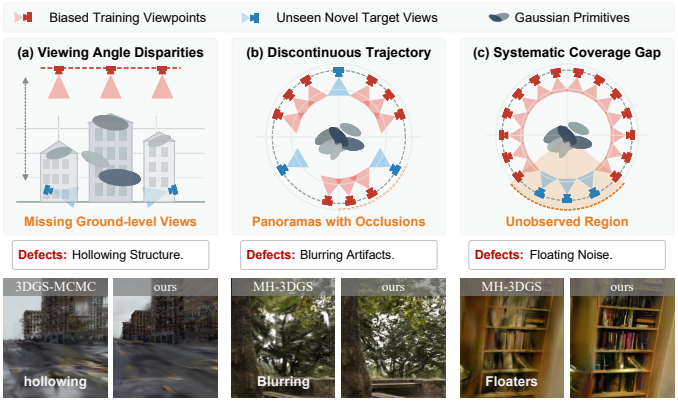


Fig. 1. Typical biased viewpoint settings. (a) Viewing angle disparities: 3DGS trained only with aerial-level views struggles to render consistent ground-level perspectives. (b) Discontinuous camera trajectories: unevenly sampled camera paths lead to fragmented scene reconstruction with blurring artifacts in synthesized views. (c) Systematic coverage gaps: missing observations in specific regions leave parts of the scene weakly supervised.

However, obtaining full-range data coverage is often impractical in real-world scenarios due to limited hardware accessibility, physical restrictions and environmental factors. Common examples include top-down drone imagery with missing ground-level views, or street-level panoramas with blind spots and occlusions. These limitations often lead to *biased viewpoints*, which manifest as viewing angle disparities, discontinuous camera trajectories and systematic coverage gaps, as shown in Fig. 1.

Under biased viewpoints, most existing 3DGS methods struggle to generalize beyond the observed perspectives. Since their optimizations are dominated by per-frame photometric constraints, Gaussian primitives tend to overfit to the biased observations, leading to suboptimal scene structure and limited global consistency. We refer to this phenomenon as *perspective overfitting*. As a result, the synthesized views from unseen viewpoints often exhibit severe artifacts, including ghosting, floating noise, and hollowing effects, limiting the immersive experience during free navigation with 6 Degrees of Freedom.

Recent efforts attempt to alleviate this issue via additional regularization or stochastic exploration. Regularization-based approaches such as depth regularization [2]–[4] are still tightly coupled to pixel-wise supervision from biased viewpoints, leading to fragmented geometry where Gaussian primitives are dense in observed regions but sparse in unseen areas. In addition, the oversmoothed depth estimation can further introduce artifacts such as floating noise and blurring. Al-

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ternatively, MCMC-based methods [5], [6] introduce stochasticity into 3DGS optimization, improving exploration around the photometric gradients. However, this stochasticity is still primarily guided by photometric constraints with only hand-crafted rules, leading to limited global exploration around the view-dependent gradients and insufficient correction toward scene-level consistency.

Fundamentally, synthesizing faithful novel views from biased viewpoints is an ill-posed problem. In weakly observed regions, the photometric supervision is under-determined, where Gaussian primitives fit the available views equally well but may disagree on the unseen ones. As a result, the per-frame optimization, which minimizes the training loss but violates scene-level coherence, collapses onto globally inconsistent structures, causing severe view-specific overfitting artifacts. A practical solution to the biased viewpoint problem should not invent unobserved scene structure from biased observations, but should regularize the under-determined optimization toward a more consistent prior. This requires a shift from purely local photometric constraints to a scene-level consistency criterion that constrains the Gaussian primitives wherever the per-view supervision is insufficient.

In this work, we propose GS-LSS, a Learned Stochastic Sampling (LSS) framework for Gaussian Splatting that augments the standard photometric objective with observed-support regularized stochastic optimization. Specifically, our method minimizes a one-sided observed-support penalty between Gaussian primitives and an aggregated support field, with a deterministic drift that pulls weakly constrained primitives toward reliable support and an adaptive stochasticity that explores the under-determined space. To instantiate the support field, we introduce Continuous Probability Field (CPF), a data-adaptive consistency prior that aggregates geometric evidence from all training views into a continuous structural field, and predicts whether a given Gaussian primitive is consistent with this aggregated observed support. Aligning the discrete Gaussians to this field provides explicit observed-support guidance for Gaussian relocation and reshaping in weakly observed regions, while an observation-aware gating mechanism prevents over-regularization in well-observed areas. Unlike MCMC-based methods [5], [6], which inject hand-crafted stochasticity into the optimization but remain primarily driven by local photometric residuals, GS-LSS introduces explicit scene-aware deterministic and stochastic updates that concentrate Gaussian relocation and exploration where biased supervision is weak. CPF is used as an observed-support stabilizer and is not claimed to reconstruct completely unobserved geometry. Extensive experiments on three typical biased viewpoint scenarios show that GS-LSS improves scene consistency and rendering quality for unseen viewpoints compared with the evaluated Gaussian Splatting baselines. The main contributions of this work are summarized as follows:

- We investigate the behavior of representative 3DGS methods under various biased viewpoint scenarios, showing that perspective overfitting is a recurring issue caused by the under-determination of the per-frame optimization in weakly observed regions.
- We propose GS-LSS, an observed-support regularized LSS framework for 3DGS that mitigates perspective overfitting by complementing view-dependent supervision with per-primitive deterministic drift and adaptive stochastic perturbation.
- We propose a Continuous Probability Field (CPF) that serves as a data-adaptive consistency prior, approximating aggregated observed support as a smooth and continuous field for Gaussian relocation and reshaping.

II. Related Work

We review recent 3DGS improvements in scene consistency and rendering quality along three directions: Gaussian Splatting and its Variants, Regularized and Guided 3DGS Optimization, and Exploring Global Consistency in 3DGS. , and discuss why biased viewpoints remain challenging.

A. Gaussian Splatting and its Variants

3D Gaussian Splatting (3DGS) [1] has emerged as a fast alternative to Neural Radiance Fields [7], building on volume rendering principles [8] with an efficient splatting representation. Subsequent work mitigates aliasing with band-limit filtering [9], reduces memory through compact representations [10], [11], improves convergence with redefined densification [5], [12]–[14], and adapts 3DGS to tasks such as sketch-driven face editing [15] and relightable rendering [16].

Nevertheless, the optimization scheme of these methods is primarily driven by view-dependent photometric supervision. Under biased viewpoint scenarios, this behavior can lead to perspective overfitting, where Gaussian primitives converge to biased observations rather than a view-consistent scene support. As a result, rendering from unseen viewpoints often suffers from visual degradation and inconsistency, exhibiting severe artifacts such as ghosting, hollowing, and floating noise.

B. Regularized and Guided 3DGS Optimization

To alleviate the limitations of purely photometric supervision, various regularization techniques have been introduced to improve the geometric fidelity and cross-view coherence, including depth regularization, generative enhancement, and multi-view regularization.

Depth regularization is a common approach to enforce scene geometry fidelity. For instance, DRGS [17], PGSR [2] and Hierarchical-GS [3] incorporate depth estimation priors, such as Depth Anything [18], to supervise the scene geometry represented by Gaussian primitives. Similar approaches have been adopted in large-scale scene reconstruction [3], [19], where both estimated depth priors and normal maps are leveraged to improve the geometric accuracy and view consistency. Although depth regularization can improve scene geometry, it is still driven by the pixel-wise depth supervision from biased viewpoints. As a result, the reconstructed scene often becomes fragmented, with Gaussian primitives located mainly in the observed regions. Moreover, the inaccuracies and oversmoothing in learned priors can further introduce artifacts including floating noise and blurring.

Another guided method uses generative priors to provide direct enhancement for synthetic views. For example, Difix3D [20] leverages diffusion model to refine the visual quality of the renderings. However, excessive reliance on uncontrollable generative priors can introduce severe hallucination artifacts, which are undesirable in many applications.

Multi-view regularization is another effective approach to enforce cross-view consistency. For example, PGSR [2] introduces geometric consistency constraints by minimizing the reprojection error of Gaussian primitives across neighboring views, while MVGS [21] further extends this approach by introducing cross-intrinsic guidance to handle more distant views. By contrast, feed-forward multi-view methods, such as SplatFormer [22], employ transformer-based deep networks to process multiple views simultaneously, but require excessive computational resources, thus limiting their practical applicability. Although multi-view regularization improves local consistency of neighboring views, it remains driven by the available training samples. Under severely biased viewpoints, the optimization stays trapped in the observed regions, resulting in limited improvement for unseen viewpoints.

C. Exploring Global Consistency in 3DGS

Apart from the regularization-based methods, another line of research focuses on improving the global consistency of 3DGS by introducing stochastic exploration mechanisms.

Stochastic sampling methods [5], [6], [23], [24] aim to improve global consistency by introducing stochasticity into the optimization, allowing Gaussian primitives to escape local minima around biased photometric gradients. 3DGS-MCMC [5] and MH-3DGS [6] reformulate Gaussian optimization as a Markov Chain Monte Carlo (MCMC) process, adding random perturbation to Gaussian locations to encourage broader exploration. Subsequent methods, including DBS [24] and Accelerating [23], further adapt stochasticity to the optimization of other Gaussian attributes, such as covariance and opacity, to enhance the global coherence of Gaussian shapes and appearances. These methods have demonstrated that introducing stochasticity into 3DGS optimization improves robustness and global consistency. However, their stochasticity is typically computed from Gaussian states that remain strongly controlled by the photometric gradients. With the limited flexibility of hand-crafted rules, the stochasticity provides only limited exploration around the photometric gradients, which is insufficient to regularize the Gaussians toward a coherent scene support under severely biased observations.

In this work, we propose a Learned Stochastic Sampling (LSS) framework that augments view-dependent photometric supervision with CPF-guided observed-support regularization, providing a practical consistency criterion that regularizes Gaussian primitives toward a less biased scene representation and mitigates perspective overfitting under biased viewpoints.

III. Method

Our approach addresses the biased observation problem by introducing observed-support regularized Learned Stochastic Sampling. The overall workflow is illustrated in Fig. 2.

A. Preliminary

Let $\mathcal{G} = \{\mathbf{g}_i\}_{i=1}^{N}$ denote the Gaussian field, $\mathcal{I} = \{\mathbf{I}_j\}_{j=1}^{M}$ denote the training view distribution where $\mathbf{I}_j$ is the j-th training view, $\mathcal{S}$ denote the reliable scene support, and $\mathcal{P}(\mathcal{G}|\mathcal{S})$ denote the posterior distribution of the Gaussian field given the $\mathcal{S}$.

3DGS-MCMC [5] reformulates Gaussian updates as a stochastic optimization process. Its update can be written in a Langevin Monte Carlo (LMC) [25] form as follows:

$$\mathcal{G}_{t+1} = \mathcal{G}_t + \alpha \cdot \nabla_{\mathcal{G}_t} \log \mathcal{P}(\mathcal{G}_t|\mathcal{S}) + \beta \cdot \epsilon_{\mathcal{G}_t|\mathcal{S}}, \tag{1}$$

where the gradient term provides a deterministic drift and $\epsilon_{\mathcal{G}_t|\mathcal{S}}$ enables stochastic exploration to escape local minima.

In practice, 3DGS-MCMC [5] approximates this update primarily through the photometric loss, which is reliable only when the viewpoint distribution $\mathcal{I}$ sufficiently covers the scene support.

$$\mathcal{P}(\mathcal{G}|\mathcal{S}) \propto \exp\left(-\mathcal{L}_{\mathbf{I}}\right), \tag{2}$$

$$\mathcal{L}_{\mathbf{I}} = (1-\eta_{\mathbf{I}}) \cdot \mathcal{L}_1 + \eta_{\mathbf{I}} \cdot \mathcal{L}_{ssim} + \eta_o \cdot |o| + \eta_{\mathbf{\Sigma}} |\sqrt{\text{eig}(\mathbf{\Sigma})}|, \tag{3}$$

where $\mathbf{I} \sim \mathcal{I}$ are the training views; $\mathcal{L}_{\mathbf{I}}$ is the pixel-wise loss function; $\text{eig}(\cdot)$ denotes the eigenvalues of the covariance matrix; η terms are hyperparameters. The stochastic perturbation σ_{mc}, which samples around the photometric gradients, is defined exclusively on the locations of the Gaussians with

$$\sigma_{mc} = \frac{\lambda_{lr} \cdot \mathbf{\Sigma}}{1 + e^{100 \cdot (0.995 - \mathbf{o})}}. \tag{4}$$

Although 3DGS-MCMC [5] improves the robustness of view synthesis through stochastic optimization, its practical update is still dominated by photometric gradients, and its hand-crafted stochasticity provides only limited exploration around the optimization bias. Therefore, we approximate an aggregated observed-support field and use it to regularize the stochastic Gaussian optimization process.

B. Learned Stochastic Sampling

1) Problem Statement: The key assumption behind the 3DGS-MCMC [5] stochastic optimization is that the viewpoint distribution $\mathcal{I}$ sufficiently covers the reliable scene support $\mathcal{S}$. However, in biased viewpoint scenarios, the training observations represent a severely *biased sampling* of the ideal scene support,

$$\mathcal{I}_{\text{bias}} \neq \mathcal{I}_{\text{ideal}} \sim \mathcal{S}. \tag{5}$$

where $\mathcal{I}_{\text{bias}}$ and $\mathcal{I}_{\text{ideal}}$ denote the biased and ideal viewpoint distributions, respectively. As a result, minimizing the photometric loss alone from biased samples cannot enforce reliable scene consistency. We refer to this phenomenon as *perspective overfitting*. One way to quantify this mismatch is the gap Δ_{overfit} between the expected loss under ideal and biased viewpoint distributions,

$$\Delta_{\text{overfit}} = \mathbb{E}_{\mathbf{I} \sim \mathcal{I}_{\text{ideal}}} \left[\mathcal{L}_{\mathbf{I}}(\mathcal{G}; \mathbf{I})\right] - \mathbb{E}_{\mathbf{I} \sim \mathcal{I}_{\text{bias}}} \left[\mathcal{L}_{\mathbf{I}}(\mathcal{G}; \mathbf{I})\right], \tag{6}$$

where $|\Delta_{\text{overfit}}|$ is a conceptual measure expected to grow with the bias severity of the viewpoint coverage, so the photometric gradients derived from severely biased observations can become poorly aligned with reliable scene support.

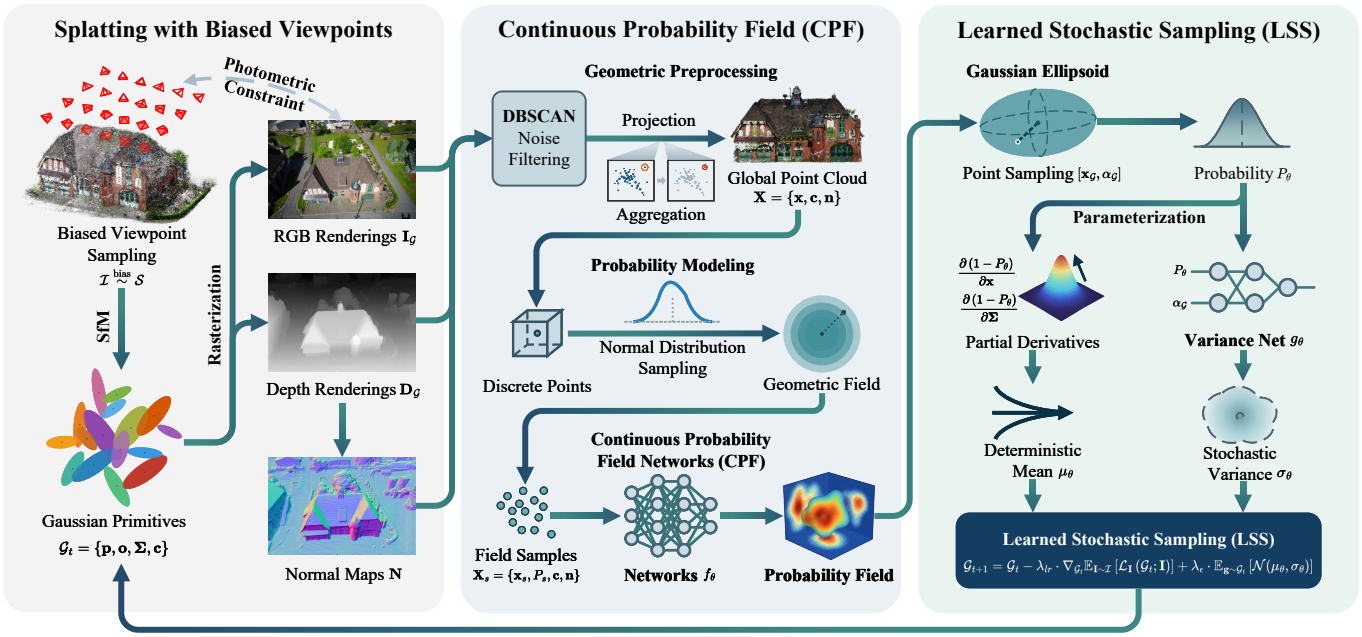


Fig. 2. The pipeline of our proposed GS-LSS. We optimize Gaussian primitives as a learned stochastic sampling process, where the deterministic and stochastic parameters are learned from the Continuous Probability Field (CPF). The CPF is trained from a probabilized geometry field, which includes points, colors, normals, and corresponding probabilities.

Addressing this challenge requires a direct constraint that regularizes the Gaussian field $\mathcal{G}$ toward reliable observed support, without assuming that fully unobserved structure can be recovered from biased views. Therefore, we define the following observed-support regularized surrogate objective,

$$\pi^\star(\mathcal{G}) \propto \exp\left(-\mathbb{E}_{\mathbf{I}\sim\mathcal{I}_{\text{bias}}}[\mathcal{L}_\mathbf{I}(\mathcal{G};\mathbf{I})] - \lambda_s \mathcal{R}(\mathcal{G};\mathcal{S})\right), \quad (7)$$

where $\mathcal{R}(\cdot)$ is a one-sided observed-support penalty and λ_s balances the photometric loss and the support regularization. Eq. 7 should be understood as a practical optimization objective rather than a claim that the full scene support is exactly recovered.

2) Parameterization of μ_θ: Assuming an ideal support field $\mathcal{S}$ is given, we then rewrite the corresponding surrogate energy as,

$$-\log \pi^\star(\mathcal{G}) \triangleq \underbrace{\mathbb{E}_{\mathbf{I}\sim\mathcal{I}}[\mathcal{L}_\mathbf{I}(\mathcal{G};\mathbf{I})]}_{\text{view-dependent}} + \lambda_s \underbrace{\mathcal{R}(\mathcal{G};\mathcal{S})}_{\text{observed-support reg}} + C, \quad (8)$$

where C is a constant independent of $\mathcal{G}$. The view-dependent term preserves faithfulness to the training views, while the support term regularizes the Gaussian field toward reliable observed support. Substituting Eq. 8 into Eq. 1 gives,

$$\begin{aligned}\mathcal{G}_{t+1} =& \mathcal{G}_t - \alpha \cdot \nabla_{\mathcal{G}_t} \mathbb{E}_{\mathbf{I}\sim\mathcal{I}_{\text{bias}}}[\mathcal{L}_\mathbf{I}(\mathcal{G}_t;\mathbf{I})] \\ & - \alpha\lambda_s \cdot \nabla_{\mathcal{G}_t} \mathcal{R}(\mathcal{G}_t;\mathcal{S}) + \beta \cdot \epsilon_{\mathcal{G}_t|\mathcal{S}}.\end{aligned} \quad (9)$$

To regularize the Gaussian field $\mathcal{G}$ with the support field $\mathcal{S}$, we define the one-sided observed-support penalty $\mathcal{R}(\cdot)$ as the expectation over Gaussian primitives,

$$\mathcal{R}(\mathcal{G};\mathcal{S}) = \mathbb{E}_{\mathbf{g}\sim\mathcal{G}}\left[\mathbf{1} - \mathcal{S}(\mathbf{g})\right], \quad (10)$$

where $\mathcal{S}(\mathbf{g}) \in [0,1]$ denotes the ideal support probability, i.e., the compatibility of Gaussian $\mathbf{g}$ with reliable scene support.

In practice, $\mathcal{S}$ is often unavailable under biased viewpoints. Therefore, we use a neural network to estimate an observed support surrogate from in-training geometric priors. We name this model Continuous Probability Field (CPF), denoted f_θ,

$$\mathcal{S}(\mathbf{g}) \approx f_\theta(\mathbf{g}), \quad (11)$$

where $f_\theta(\mathbf{g})$ predicts the probability of a given 3D primitive being supported by the aggregated observed scene structure. A larger deviation of the Gaussian primitive from this observed support corresponds to a lower probability.

Now, we can obtain a tractable observed-support penalty:

$$\widehat{\mathcal{R}}(\mathcal{G};\mathcal{S}) = \mathbb{E}_{\mathbf{g}\sim\mathcal{G}}\left[1 - f_\theta(\mathbf{g})\right]. \quad (12)$$

Let $\delta(\mathbf{g}) = f_\theta(\mathbf{g}) - \mathcal{S}(\mathbf{g})$ denote the CPF approximation error. Then, we have

$$\nabla_{\mathcal{G}_t}\widehat{\mathcal{R}}(\mathcal{G}_t;\mathcal{S}) - \nabla_{\mathcal{G}_t}\mathcal{R}(\mathcal{G}_t;\mathcal{S}) = \nabla_{\mathcal{G}_t}\mathbb{E}_{\mathbf{g}\sim\mathcal{G}_t}\left[\delta(\mathbf{g})\right], \quad (13)$$

This term measures the local sensitivity of the CPF, not a convergence guarantee. Replacing the ideal support $\mathcal{S}$ with CPF changes the support gradient direction by the residual drift in Eq. 13. When the aggregated priors are reliable, the drift can be reduced and the CPF mainly stabilizes weakly observed Gaussians. The global aggregation, delayed start, normal smoothness, DBSCAN filtering, and observation-aware gating in Section III-C are therefore safeguards that reduce, but do not eliminate, CPF error.

By replacing $\mathcal{R}(\cdot)$ with $\widehat{\mathcal{R}}(\cdot)$ in Eq. 9, we have,

$$\begin{aligned}\mathcal{G}_{t+1} =& \mathcal{G}_t - \alpha \cdot \nabla_{\mathcal{G}_t} \mathbb{E}_{\mathbf{I}\sim\mathcal{I}_{\text{bias}}}[\mathcal{L}_\mathbf{I}(\mathcal{G}_t;\mathbf{I})] \\ & - \alpha\lambda_s \cdot \nabla_{\mathcal{G}_t} \mathbb{E}_{\mathbf{g}\sim\mathcal{G}_t}\left[1 - f_\theta(\mathbf{g})\right] + \beta \cdot \epsilon_{\mathcal{G}_t|\mathcal{S}}.\end{aligned} \quad (14)$$

Therefore, the learned deterministic drift is defined as,

$$\mu_\theta = -\nabla_\mathbf{g}(1 - f_\theta(\mathbf{g})). \quad (15)$$

The parameterization of μ_θ provides explicit observed-support guidance for the Gaussians by regularizing them toward the observed-support field.

3) Parameterization of σ_θ: The stochasticity of the LMC sampling process is crucial for global exploration. Under biased viewpoints, a good perturbation should depend on both the observed support field and the current Gaussian states. In addition, as discussed in [26], the variance should be adaptive to the training state. Therefore, we propose to parameterize $\epsilon_{\mathcal{G}_t|\mathcal{S}}$ with a variance network g_θ that takes the predicted probability from f_θ and the Gaussian field $\mathcal{G}_t$ as input,

$$\epsilon_{\mathcal{G}_t|\mathcal{S}} = \mathcal{N}(0, \sigma_\theta), \sigma_\theta = g_\theta\big(f_\theta(\mathcal{G}_t); \mathcal{G}_t\big). \tag{16}$$

Compared with the hand-crafted stochasticity in 3DGS-MCMC [5], the parameterization of σ_θ is conditioned on both the observed support field and the Gaussian states. When Gaussian primitives are far from the observed support, the low probability predicted by the CPF model yields a large variance, encouraging stronger exploration to escape local minima.

4) Final Formulation: Combining Eq. 15 and Eq. 16, we implement GS-LSS as a per-primitive reparameterized stochastic update. For primitive i, let $\boldsymbol{\tau}_i = [\mathbf{p}_i, \boldsymbol{\rho}_i, \mathbf{q}_i]$ denote the center, log-scale, and normalized-rotation parameters defined in Sec. III-C. The update is

$$\begin{aligned}\boldsymbol{\tau}_i^{t+1} &= \boldsymbol{\tau}_i^t - \lambda_{lr}\nabla_{\boldsymbol{\tau}_i}\mathcal{L}_{\mathbf{I}} + \lambda_s\mu_{\theta,i} \\ &\quad + \lambda_\epsilon\sigma_{\theta,i}\odot\boldsymbol{\xi}_i, \quad \boldsymbol{\xi}_i \sim \mathcal{N}(0, \mathbf{I}), \end{aligned}\tag{17}$$

$$\mu_{\theta,i} = \nabla_{\boldsymbol{\tau}_i} f_\theta(\mathbf{x}_{\mathcal{G},i}), \tag{18}$$

$$\sigma_{\theta,i} = g_\theta\big(f_\theta(\mathbf{x}_{\mathcal{G},i}), \alpha_{\mathcal{G},i}; \boldsymbol{\tau}_i\big). \tag{19}$$

This form makes the role of $\sigma_{\theta,i}$ explicit: it scales the sampled noise rather than disappearing inside an expectation. We use the stochastic formulation as a learned optimizer rather than an exact Langevin or Metropolis-Hastings sampler, and therefore do not claim a stationary sampling distribution.

The effectiveness of μ_θ and σ_θ depends on the accuracy of the observed support approximation by the CPF model. Therefore, the training priors and loss functions for the CPF model are crucial for the performance of our LSS method.

C. Continuous Probability Field

The target of the CPF model f_θ is to provide a data-driven approximation of the aggregated observed-support field $\mathcal{S}(\mathbf{g})$ (Eq. 11). Operationally, the CPF probability P_θ is queried at points sampled from Gaussian primitives, which denotes observed-support probability rather than a posterior over unseen geometry. The accuracy of the CPF model is reflected in the residual drift in Eq. 13. To construct this approximation from biased training samples, we supervise f_θ with an in-training probabilized geometry field aggregated from all training views.

The geometry that supervises the CPF model is itself derived from 3DGS under biased viewpoints, so the CPF model is not claimed to recover fully unobserved geometry. Instead, it estimates a smooth observed-support field from evidence aggregated across all available training views, and uses this field to regularize updates only where per-view supervision is under-determined. In our training schedule, CPF supervision starts after the initial Gaussian field has stabilized, and learned LSS sampling is enabled only after the CPF has received its first supervision stage. The global aggregation, DBSCAN outlier filtering, normal-driven smoothness, and observation-aware gating reduce the risk that a single biased view dominates the field.

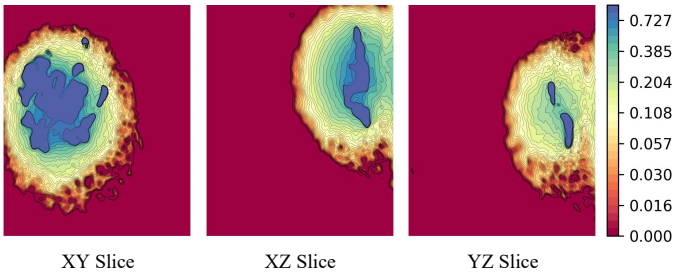


Fig. 3. A visual example of a trained Continuous Probability Field (CPF). The panels show $x-y$, $x-z$, and $y-z$ slices, blue and red denote high and low observed-support probability, respectively.

1) In-training Geometric Field: Given Gaussian ellipsoids $\mathcal{G}_t = \{\mathbf{g}_i\}_{i=1}^N$, where $\mathbf{g}_i = \{\mathbf{p}, \mathbf{o}, \boldsymbol{\Sigma}, \mathbf{c}\}$ at training iteration t, we first render RGB images $\mathbf{I}_\mathcal{G}$ and depth maps $\mathbf{D}_\mathcal{G}$ from the training views, and compute the normal maps $\mathbf{N}$ from the depth maps. After DBSCAN outlier filtering, we project these maps and aggregate them into the world coordinate system to construct a global sparse point cloud $\mathbf{X} = \{\mathbf{x}, \mathbf{c}, \mathbf{n}\}$, where $\mathbf{x}$ is the 3D coordinate, $\mathbf{c}$ is color, and $\mathbf{n}$ is normal.

To obtain probabilistic supervision without making the field dependent on the absolute scene scale, we first normalize the scene coordinates by the scene center $\mathbf{c}_{\text{sc}}$ and diagonal length d_{sc}, i.e., $\tilde{\mathbf{x}} = (\mathbf{x} - \mathbf{c}_{\text{sc}})/d_{\text{sc}}$. We then sample around each normalized point and compute its probability,

$$\tilde{\mathbf{x}}_s \sim \mathcal{N}(\tilde{\mathbf{x}}, \sigma_s^2\mathbf{I}_3), \tag{20}$$

$$P_s = \exp\left(-\frac{\|\tilde{\mathbf{x}}_s - \tilde{\mathbf{x}}\|_2^2}{2\sigma_s^2}\right), \tag{21}$$

where σ_s is set proportional to the median nearest-neighbor distance of the aggregated point cloud in normalized coordinates. We thus acquire the geometry field $\mathbf{X}_s = \{\tilde{\mathbf{x}}_s, P_s, \mathbf{c}, \mathbf{n}\}$ as the training priors. This normalization makes the CPF supervision comparable across datasets with different physical scales.

2) CPF network and Loss Function: We build the CPF network f_θ with a 3-layer SIREN [27]-style MLP, which maps sampled points $\mathbf{x}_s$ to the probability P_θ and color $\mathbf{c}_\theta$. During training, the depth normals are consistently oriented before aggregation, and the normalized CPF gradient $\hat{\mathbf{n}}_\theta = \nabla_\mathbf{x} P_\theta / (\|\nabla_\mathbf{x} P_\theta\|_2 + \epsilon_n)$ is regularized by $\mathbf{n}$ to encourage field continuity and smoothness. The loss function is defined as:

$$\begin{aligned}\mathcal{L}_{cpf} = &\sum_\mathbf{x} |P_\theta - P_s| + \sum_\mathbf{x} |1 - \langle \mathbf{n}, \nabla_\mathbf{x} P_\theta \rangle| \\ &+ \sum_\mathbf{x} |\mathbf{c}_\theta - \mathbf{c}| \end{aligned}\tag{22}$$

where $\langle \cdot, \cdot \rangle$ denotes cosine similarity. Fig. 3 shows an example of the trained CPF model.

Since the loss function is built upon globally aggregated geometric priors, it encourages the CPF network f_θ to behave as an observed-support estimator rather than a per-view regressor, which is designed to reduce the residual drift $\nabla_{\mathcal{G}_t}\mathbb{E}_{\mathbf{g}\sim\mathcal{G}_t}[\delta(\mathbf{g})]$ in Eq. 13 when the priors are reliable. This is the practical distinction from regularization-based methods, whose regularization is derived from per-view perspectives.

3) Computation of μ_θ and σ_θ: To make the LSS update of Eq. 17 concrete, we compute the deterministic drift $\mu_{\theta,i}$ and the stochastic scale $\sigma_{\theta,i}$ as explicit, differentiable functions of the CPF predictions and the Gaussian states. Different from 3DGS-MCMC [5], we update both the locations and shapes of the Gaussian ellipsoids through the gradients provided by the CPF model. For each Gaussian ellipsoid $\mathbf{g}_i = \{\mathbf{p}_i, \mathbf{o}_i, \mathbf{\Sigma}_i, \mathbf{c}_i\} \sim \mathcal{G}$, we randomly sample a point within it and compute its opacity,

$$\mathbf{x}_{\mathcal{G},i} \sim \mathcal{N}\left(\mathbf{p}_i, \mathbf{\Sigma}_i\right), \tag{23}$$

$$\alpha_{\mathcal{G},i} = e^{-\frac{1}{2}(\mathbf{x}_{\mathcal{G},i}-\mathbf{p}_i)^T\mathbf{\Sigma}_i^{-1}(\mathbf{x}_{\mathcal{G},i}-\mathbf{p}_i)} \cdot \mathbf{o}_i. \tag{24}$$

The CPF then gives the per-Gaussian support probability $P_{\theta,i} = f_\theta(\mathbf{x}_{\mathcal{G},i})$.

In implementation, we do not directly update the raw covariance matrix. Instead, the covariance is parameterized as $\mathbf{\Sigma}_i = \mathbf{R}(\mathbf{q}_i)\,\mathrm{diag}(\exp(2\boldsymbol{\rho}_i))\mathbf{R}(\mathbf{q}_i)^T$, where $\boldsymbol{\rho}_i$ are log-scales and $\mathbf{q}_i$ is a normalized quaternion. Therefore, the LSS geometry parameter is $\boldsymbol{\tau}_i = [\mathbf{p}_i, \boldsymbol{\rho}_i, \mathbf{q}_i]$, which preserves the positive definiteness of $\mathbf{\Sigma}_i$ during optimization. Following Eq. 18, the deterministic drift is obtained by back-propagating the CPF probability to these parameters:

$$\mu_{\theta,i} = -\nabla_{\boldsymbol{\tau}_i}(1 - P_{\theta,i}) = \nabla_{\boldsymbol{\tau}_i}P_{\theta,i}. \tag{25}$$

This drift moves each Gaussian toward higher support $P_{\theta,i}$ while keeping the covariance representation valid.

The variance module is implemented with two lightweight 2-layer MLP heads. The scale head g_θ predicts a learnable scale $a_{\theta,i}$, and the gate head h_θ predicts an observation-aware confidence gate $\Delta_{\theta,i}$ from the observed support $P_{\theta,i}$ and sampled opacity $\alpha_{\mathcal{G},i}$:

$$a_{\theta,i} = g_\theta\left(P_{\theta,i}, \alpha_{\mathcal{G},i}\right), \quad \Delta_{\theta,i} = h_\theta\left(P_{\theta,i}, \alpha_{\mathcal{G},i}\right), \tag{26}$$

$$\sigma_{\theta,i} = \frac{a_{\theta,i} \cdot \mathbf{D}_i}{1 + e^{100\cdot\Delta_{\theta,i}}}, \tag{27}$$

where $\mathbf{D}_i$ is a diagonal parameter-space scale matrix: its position block uses the current Gaussian axis scales $\exp(\boldsymbol{\rho}_i)$, while the log-scale and rotation blocks use fixed small constants. P_θ measures support from the CPF field, and $\alpha_{\mathcal{G}}$ measures primitive support at the sampled point. The learned gate suppresses $\sigma_{\theta,i}$ for well-observed primitives and allows larger perturbations for weakly supported primitives. The factor 100 follows the sharp confidence gate used in 3DGS-MCMC [5]. We clamp the exponent for numerical stability. The heads g_θ and h_θ are not trained with a separate variance target. After the one-step reparameterized update in Eq. 17, Eq. 28 is back-propagated through $\mu_{\theta,i}$ and $\sigma_{\theta,i}$ using first-order gradients, without multi-step unrolling or second-order derivatives. Thus, $\sigma_{\theta,i}$ is a train-time perturbation scale rather than a calibrated posterior variance, replacing the hand-crafted opacity rule in Eq. 4 with a learned observation-conditioned gate.

The LSS process is then performed using Eq. 17, where $\mu_{\theta,i}$ serves as an observed-support drift and $\sigma_{\theta,i}$ as an adaptive stochastic perturbation.

Algorithm 1 CPF-guided stochastic optimization in GS-LSS

```
Initialize G from SfM points and train with L_I.
for t = 1, ..., T do
    Render RGB/depth/normal maps from training views.
    if t ≥ T_CPF then
        Aggregate geometry; apply DBSCAN; update CPF with Eq. 22.
    end if
    Update ordinary 3DGS attributes with the photometric loss.
    if t ≥ T_LSS then
        Sample x_{G,i} from LSS selected Gaussian ellipsoids.
        Query CPF, compute μ_{θ,i} by Eq. 25, compute σ_{θ,i} by Eq. 27.
        Sample ξ_i ~ N(0, I) and update τ_i with Eq. 17.
        Normalize rotations and keep covariance positive definite.
    end if
end for
Discard CPF, g_θ and h_θ for inference.
return optimized Gaussian field G.
```

D. Training Objectives

GS-LSS jointly optimizes the Gaussian field $\mathcal{G}$ and the CPF model f_θ, details are provided in Algorithm 1. Following the surrogate objective in Eq. 8, the Gaussian field is optimized by the observed-support regularized objective,

$$\mathcal{L}_\mathcal{G} = \mathbb{E}_{\mathbf{I}\sim\mathcal{I}_{\text{bias}}}\left[\mathcal{L}_\mathbf{I}(\mathcal{G};\mathbf{I})\right] + \lambda_s\,\widehat{\mathcal{R}}(\mathcal{G};\mathcal{S}), \tag{28}$$

which combines the photometric loss $\mathcal{L}_\mathbf{I}$ (Eq. 3) with the observed-support penalty $\widehat{\mathcal{R}}$ (Eq. 12). The minimization of $\mathcal{L}_\mathcal{G}$ is realized through the LSS update in Eq. 17, where the deterministic drift $\mu_{\theta,i}$ decreases the support penalty and the adaptive perturbation $\sigma_{\theta,i}$ injects scene-aware stochastic exploration. The CPF networks are optimized by the geometric-field loss $\mathcal{L}_{cpf}$ (Eq. 22), which fits f_θ to the globally aggregated probability, color and normal priors. We optimize the two objectives jointly so that observed-support estimation and Gaussian optimization remain coupled during training. Importantly, the CPF and stochastic heads are used only during training. Inference is performed with only the optimized Gaussian field $\mathcal{G}$ with standard 3DGS renderer.

IV. Experiments

A. Implementation Details

1) Datasets: We evaluate our method on 16 scenes from 5 widely used datasets, including 8 surrounding scenes from Mip-NeRF 360 [28], 2 outdoor scenes (Truck and Train) from Tanks & Temples [29], 4 large-scale scenes with planar trajectories from Mill 19 [30], and 2 urban scenes with both aerial and ground views from MatrixCity [31] and Zeche [32]. These datasets cover a wide range of scene types and viewpoint distributions, providing a comprehensive benchmark.

2) Biased Viewpoint Settings: To evaluate the robustness of our method under various biased viewpoint scenarios, we design three representative settings: *systematic coverage gap*, *discontinuous trajectory* and *viewing angle disparities*.

TABLE I
EVALUATION FOR SYSTEMATIC COVERAGE GAPS ON MIP-NERF 360 [28] AND TANKS & TEMPLES [29], WITH ANGULAR GAPS 20°, 40°, 60°, 80° AND 100°. BEST RESULTS ARE IN **BOLD** AND SECOND-BEST ARE UNDERLINED.

Method	20°			40°			60°			80°			100°		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
Mip-NeRF 360															
3DGS [1]	0.787	24.936	0.202	0.726	22.137	0.249	0.674	20.499	0.291	0.630	19.460	0.324	0.584	17.597	0.358
Scaffold-GS [10]	0.760	23.844	0.231	0.706	21.835	0.273	0.653	20.418	0.315	0.616	19.574	0.344	0.571	17.751	0.378
Hierarchical-GS [3]	0.787	25.044	0.203	0.725	22.314	0.250	0.671	20.611	0.291	0.624	19.339	0.326	0.581	17.463	0.359
MVGS [21]	0.784	24.827	0.203	0.715	22.130	0.256	0.658	20.458	0.303	0.608	19.081	0.340	0.561	17.164	0.374
PGSR [2]	0.771	24.203	0.231	0.706	21.588	0.280	0.662	20.179	0.307	0.611	18.834	0.343	0.553	16.808	0.375
3DGS-MCMC [5]	0.801	25.323	0.194	0.739	22.369	0.239	0.679	20.732	0.283	0.632	19.344	0.319	0.572	17.204	0.355
MH-3DGS [6]	0.784	25.122	0.224	0.728	22.365	0.265	0.683	20.855	0.300	0.650	20.008	0.328	0.608	18.352	0.360
GS-LSS (Ours)	**0.808**	**25.839**	**0.188**	**0.752**	**23.152**	**0.236**	**0.708**	**21.615**	**0.273**	**0.667**	**20.235**	**0.307**	**0.631**	**18.805**	**0.340**
Tanks & Temples															
3DGS [1]	0.771	20.411	0.209	0.741	19.464	0.234	0.695	18.651	0.269	0.663	17.945	0.294	0.632	17.422	0.323
Scaffold-GS [10]	0.744	19.331	0.222	0.743	19.945	0.232	0.692	18.658	0.269	0.653	17.589	0.301	0.615	17.082	0.334
Hierarchical-GS [3]	0.766	20.418	0.208	0.743	19.640	0.229	0.694	18.728	0.267	0.657	17.811	0.296	0.630	17.226	0.321
MVGS [21]	0.756	20.211	0.199	0.721	19.150	0.229	0.668	18.323	0.275	0.625	17.549	0.309	0.590	16.875	0.340
PGSR [2]	0.765	20.119	0.234	0.737	19.182	0.258	0.687	18.020	0.295	0.653	17.305	0.313	0.634	17.206	0.338
3DGS-MCMC [5]	0.782	21.026	0.202	0.754	20.018	0.220	0.702	18.973	0.257	0.665	18.014	0.285	0.632	17.406	0.317
MH-3DGS [6]	0.754	19.967	0.221	0.726	19.205	0.241	0.649	17.404	0.291	0.652	17.299	0.298	0.612	16.627	0.331
GS-LSS (Ours)	**0.787**	**21.451**	**0.191**	**0.757**	**20.253**	**0.210**	**0.709**	**19.345**	**0.247**	**0.672**	**18.113**	**0.274**	**0.643**	**17.818**	**0.304**

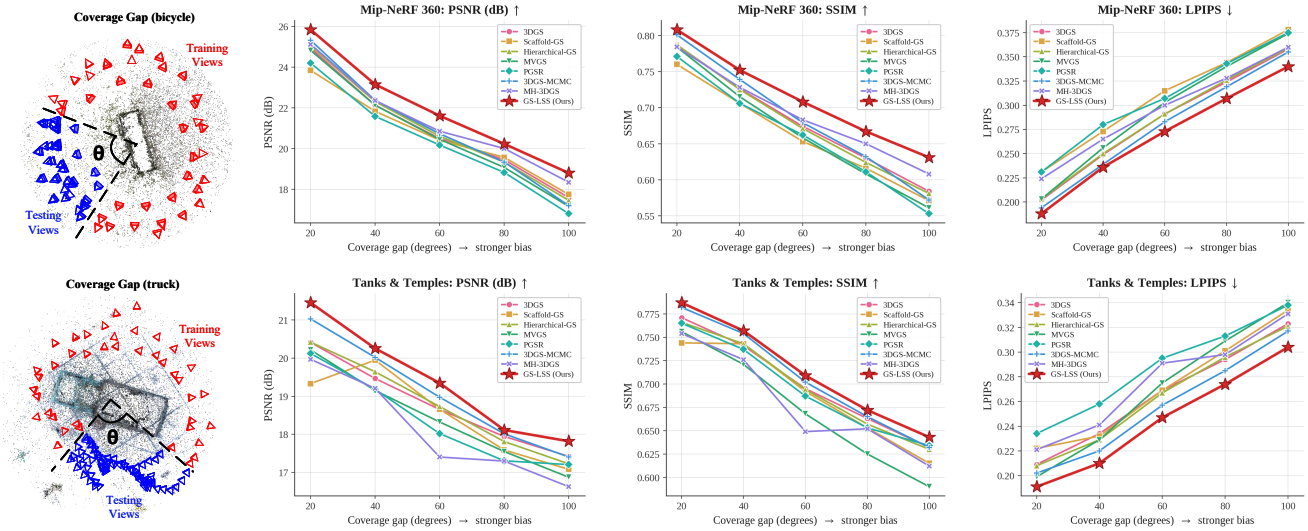


Fig. 4. Visualization of the performance trends under systematic coverage gaps, with gap size ranging from 20° to 100°. The first row corresponds to Mip-NeRF 360 [28] dataset and the second row corresponds to Tanks & Temples [29] dataset.

Systematic coverage gap. On Mip-NeRF 360 [28] and Tanks & Temples [29], we randomly remove consecutive segments of training views to create angular coverage gaps, where the removed viewpoints are used for evaluation. The size of the angular gap is set to a range from 20° to 100° to simulate different severities of systematic coverage gaps.

Discontinuous trajectory. We evaluate this setting on both surrounding and planar camera trajectories using Mip-NeRF 360 [28], Tanks & Temples [29] and Mill 19 [30]. Specifically, we randomly select several disjoint regions of the viewpoint coverage to form unevenly distributed training views, where each region contains a local cluster of neighboring viewpoints.

Viewing angle disparities. For this setting, we use Block A of MatrixCity [31] and Zeche [32], as they both contain substantial collections of aerial and ground views. In our experiments, all methods are trained exclusively on aerial views and evaluated on ground views to assess robustness against large view angle disparities. For all settings, the train/test split is applied before SfM-based initialization. The sparse points for Gaussian initialization, the footprint in Fig. 5, and the FFR support in Table IV are computed from the selected training views only. No held-out views are used by any method during training or support construction.

3) Competitors and Hyperparameters: We compare our method against representative 3DGS baselines from three families. Among them, 3DGS [1] and Scaffold-GS [10] are two widely used photometric optimization baselines. Hierarchical-GS [3], MVGS [21] and PGSR [2] are three representative geometric regularization methods for 3DGS. 3DGS-MCMC [5] and MH-3DGS [6] are two state-of-the-art stochastic optimiza-

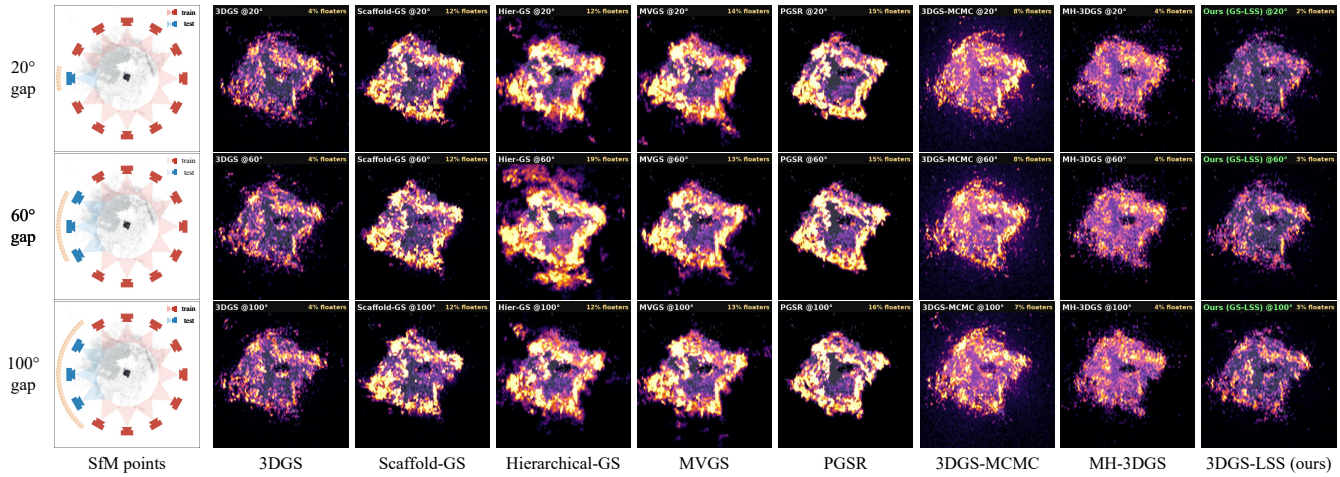


Fig. 5. Top-down bird's eye view of free-space floaters for the Bonsai scene in Mip-NeRF 360 [28] under different systematic coverage gap settings. The faint gray shows the SfM footprint, and the hot color map shows the relative density of Gaussians identified as free-space floaters by the visualization protocol.

tion baselines. Particularly, for Hierarchical-GS [3], we enable only depth regularization to ensure a fair comparison. Methods that require explicit control over the number of Gaussian primitives use the same total count across all competitors.

All baselines are trained with their official implementations and default configurations under the same training iteration budget, image resolution, fixed train/test split, and the same initialization point cloud. Scaffold-GS uses its official anchor and view-adaptive rendering settings. For our method, CPF training begins at iteration 2000 with a learning rate of 1×10^{-4}, and the learned sampling of μ_θ and σ_θ begins at iteration 2500. Unless otherwise stated, the photometric loss of GS-LSS is computed from one randomly sampled training view per iteration, following the standard 3DGS-MCMC [5] protocol. Other hyperparameters are set as follows: CPF net f_θ is a 3-layer SIREN-style MLP, g_θ and h_θ are 2-layer MLPs; DBSCAN $\epsilon = 0.25$ in normalized coordinates; $\lambda_s = 0.1$; and $\lambda_\epsilon = 1$. All experiments are conducted on a single NVIDIA RTX 4090 GPU with 24 GB memory.

B. Main Results

We organize the experiments to test our core claims: 1) performance degradation under biased viewpoints is consistent with perspective overfitting. 2) hand-crafted stochastic exploration alleviates this issue, but remains limited without an explicit observed-support guidance. 3) CPF-guided observed-support regularization provides an effective stabilizer for 3DGS optimization under biased viewpoints.

1) Claim 1: Biased viewpoints induce perspective overfitting: Table I and Fig. 4 summarize the quantitative results for the systematic coverage gap setting. These results show that all methods degrade substantially as the coverage gap widens, which indicates that the deterioration is not a dataset-specific or method-specific phenomenon, but a general issue for 3DGS under biased viewpoints. Geometric regularization methods, including Hierarchical-GS [3], MVGS [21] and PGSR [2], also remain vulnerable under large gaps. This indicates that observation-driven regularization constraints, such as depth supervision and multi-view reprojection, remain tightly coupled to the available viewpoints. As a result, the learned structures are locally consistent with the observed perspectives while leaving the unseen regions weakly constrained. The qualitative comparisons in Fig. 6 support this explanation. The rendered images from MVGS [21] and PGSR [2] commonly exhibit significant hollowing artifacts in the gap regions, which is a typical symptom of perspective overfitting.

More direct evidence of perspective overfitting is shown in Fig. 5 and Table IV, which visualize and quantify the free-space floaters of different methods on Mip-NeRF 360 [28]. Under large coverage gaps, most methods place many spurious Gaussians in free space that are far away from the real surfaces, indicating that their optimization creates unconstrained geometry in the unobserved regions. By contrast, our GS-LSS method visibly reduces free-space floater density in the displayed gap settings, suggesting that the observed-support guidance helps suppress the unconstrained densification induced by biased sampling. This phenomenon is not limited to the systematic coverage gap setting. Under aerial-ground view disparities (Table III), where the training and testing views differ dramatically in elevation angles, geometric regularization methods commonly suffer from inferior performance compared to other methods. Hierarchical-GS [3] and PGSR [2] drop below 13.1 dB PSNR on MatrixCity [31], far worse than the photometric and stochastic baselines that all remain above 15 dB. This further suggests that perspective overfitting is a common issue under severely biased viewpoints.

Together, these results are consistent with our claim that optimization bias toward observed perspectives is a major cause of 3DGS deterioration under biased viewpoints, and that simply adding observation-driven geometric constraints is insufficient to remove this bias.

2) Claim 2: Stochasticity alone provides limited improvements: Table I, II, and III show that stochastic exploration can improve over purely photometric optimization and geometric regularization baselines. In particular, 3DGS-MCMC [5] is consistently competitive across all biased viewpoint settings, suggesting that stochastic updates help Gaussian primitives escape locally biased optimization.

TABLE II
EVALUATION FOR DISCONTINUOUS TRAJECTORIES ON MIP-NERF 360 [28], TANKS & TEMPLES [29] AND MILL 19 [30]. BEST RESULTS ARE IN **BOLD** AND SECOND-BEST ARE UNDERLINED.

Method	Mip-NeRF 360			Tanks & Temples			Mill 19		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
3DGS [1]	0.826	26.979	0.169	0.778	20.914	0.216	0.754	21.503	0.217
Scaffold-GS [10]	0.783	25.356	0.205	0.745	19.496	0.237	0.755	21.081	0.226
Hierarchical-GS [3]	0.827	27.046	0.168	0.787	21.521	0.210	0.762	22.263	0.210
MVGS [21]	0.828	27.288	0.164	0.776	20.744	0.196	0.746	21.311	0.227
PGSR [2]	0.821	26.269	0.188	0.786	21.093	0.231	0.717	20.336	0.265
3DGS-MCMC [5]	0.843	27.517	0.153	0.799	21.955	0.196	0.766	22.532	0.207
MH-3DGS [6]	0.817	26.895	0.191	0.767	20.895	0.218	0.747	22.394	0.249
GS-LSS (Ours)	**0.845**	**28.018**	**0.151**	**0.805**	**22.160**	**0.180**	**0.775**	**22.893**	**0.198**

TABLE III
EVALUATION FOR VIEWING ANGLE DISPARITIES ON MATRIXCITY [31] AND ZECHE [32] DATASETS. BEST RESULTS ARE IN **BOLD** AND SECOND-BEST ARE UNDERLINED.

Method	MatrixCity			Zeche		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
3DGS [1]	0.578	16.146	0.495	0.330	9.431	0.568
Scaffold-GS [10]	0.562	16.193	0.493	0.337	9.806	0.591
Hierarchical-GS [3]	0.458	12.943	0.532	0.341	9.646	0.577
MVGS [21]	0.541	15.610	0.490	0.340	9.494	0.581
PGSR [2]	0.417	13.072	0.531	0.261	8.499	0.538
3DGS-MCMC [5]	0.586	16.225	0.480	0.373	10.133	0.534
MH-3DGS [6]	0.560	15.297	0.502	**0.481**	12.703	0.546
GS-LSS (Ours)	**0.594**	**16.951**	**0.462**	**0.481**	**12.879**	**0.530**

However, these results also suggest that hand-crafted stochasticity alone is still limited, as the performance of MCMC-based methods often remains below GS-LSS in the reported settings, and the performance gap widens as the bias becomes more severe. Taking the systematic coverage gap as an example, on Mip-NeRF 360 [28] the PSNR advantage of GS-LSS over 3DGS-MCMC [5] grows monotonically with the gap size, widening from 0.52 dB at the mildest 20° gap to 1.60 dB at 100°. A consistent advantage is also observed on Tanks & Temples [29], where GS-LSS outperforms 3DGS-MCMC at every gap size by 0.10-0.43 dB PSNR. The limited improvement of purely stochastic methods suggests that their optimization remains strongly influenced by view-dependent photometric gradients, which limits global coherence under severe sampling bias. Results from the discontinuous trajectory setting (Table II) and the viewing angle disparity setting (Table III) further support this observation with the same trend.

The qualitative comparison in Fig. 6 leads to the same conclusion. MCMC-based methods show relatively higher rendering quality than deterministic methods, but still exhibit blurry textures, incomplete geometry and hollow structures in severely unobserved regions.

These results support our second claim that hand-crafted stochastic exploration improves robustness to biased photometric updates, but is insufficient to enforce a coherent scene structure under severe sampling bias, as the optimization remains predominantly driven by view-dependent constraints.

3) Claim 3: CPF-guided observed-support regularization improves biased-view robustness: Results across all three biased viewpoint settings are consistent with our third claim that CPF-guided observed-support regularization provides effective guidance for 3DGS optimization. For example, under systematic coverage gaps (Table I), our method achieves the best performance across gap sizes and metrics on both Mip-NeRF 360 [28] and Tanks & Temples [29]. Fig. 4 further shows that our method maintains a clear advantage as the coverage gap becomes more severe. Under discontinuous trajectories (Table II), our method obtains the best average performance across all three datasets, outperforming the second-best method 3DGS-MCMC [5] by 0.501 dB, 0.205 dB and 0.361 dB PSNR on Mip-NeRF 360 [28], Tanks & Temples [29] and Mill 19 [30] respectively. Under viewing angle disparities (Table III), our method achieves the best performance on both MatrixCity [31] and Zeche [32], improving the PSNR over the second-best baseline by 0.468 dB and 0.176 dB, respectively.

The performance rankings further support the design of GS-LSS. The improvement over deterministic-only methods, such as 3DGS [1] and Scaffold-GS [10], supports the view that photometric optimization alone is insufficient under biased viewpoints. The improvement over geometric methods, such as Hierarchical-GS [3], MVGS [21] and PGSR [2], suggests that observation-driven regularization alone may not provide sufficient scene-level guidance under biased viewpoints. The improvement over MCMC-based methods, such as 3DGS-MCMC [5] and MH-3DGS [6], indicates that stochasticity alone provides limited exploration beyond photometric gradients. Instead, CPF-guided observed-support regularization provides explicit guidance for 3DGS optimization, helping mitigate perspective overfitting and improve novel view synthesis under various biased viewpoint scenarios.

Qualitative results in Fig. 6 further support the advantages of GS-LSS. Our method produces more coherent geometry, sharper textures and fewer artifacts in unseen regions than the competitors, consistent with the reduced free-space floater density in Fig. 5.

C. Ablation Study

We conduct ablations on floater reduction, LSS parameterization, CPF construction, and stochastic optimization stability.

1) Free-space Floater Ratio: We ablate free-space floater ratio (FFR) on Mip-NeRF 360 [28] under systematic coverage gaps to examine whether GS-LSS reduces support structural

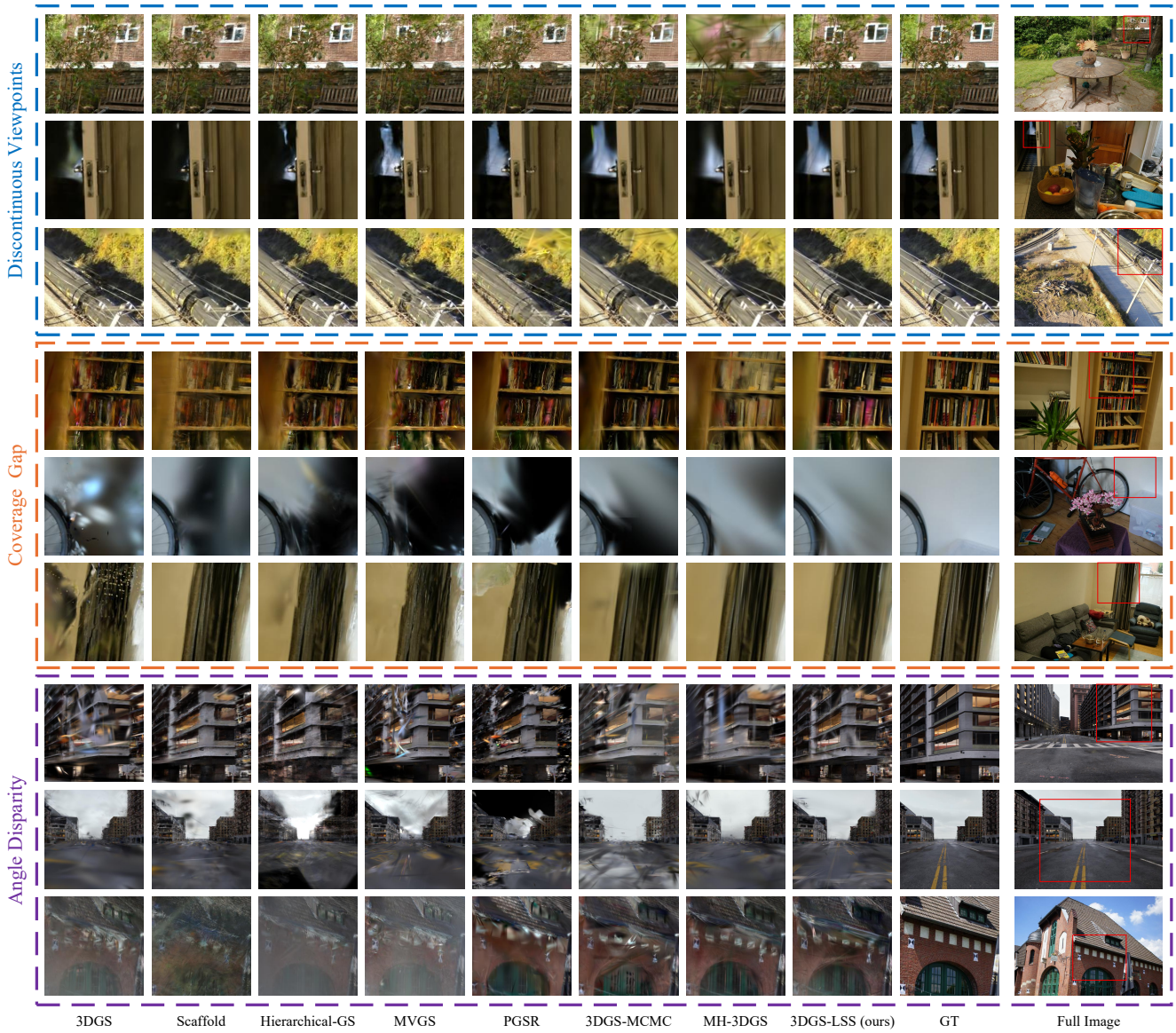


Fig. 6. Visual comparison on three biased viewpoint settings, including discontinuous trajectories, systematic coverage gaps, and viewing angle disparities.

TABLE IV
FREE-SPACE FLOATER RATIO (FFR, %) ON MIP-NERF 360 [28] UNDER SYSTEMATIC COVERAGE GAPS. LOWER IS BETTER.

Method	20°	40°	60°	80°	100°	Avg.
3DGS [1]	3.914	3.918	3.932	3.866	3.792	3.884
Scaffold-GS [10]	11.852	11.818	12.287	12.098	12.469	12.105
Hierarchical-GS [3]	12.129	12.165	18.589	12.357	11.725	13.393
MVGS [21]	13.967	13.568	13.024	13.091	12.692	13.268
PGSR [2]	14.638	14.732	15.381	14.618	15.985	15.071
3DGS-MCMC [5]	7.596	7.263	7.557	7.142	7.091	7.330
MH-3DGS [6]	3.842	4.104	3.715	3.753	3.741	3.831
GS-LSS (Ours)	**1.867**	**2.223**	**2.723**	**2.382**	**3.028**	**2.445**

floaters. FFR is defined as the opacity-weighted percentage of Gaussian centers farther than 1.5% of the scene diagonal from training-view SfM support inside the evaluated gap region. Since this CPF-independent metric is an SfM-based proxy, we use it only as a relative indicator of unsupported primitives in the weakly observed region shown in Fig. 5. Results in Table IV show that GS-LSS achieves the lowest average FFR across all gap sizes. This ablation supports the claim that CPF-guided optimization reduces unsupported free-space primitives, while Table V further identifies which update and observed-support components contribute to this behavior.

TABLE V
ABLATION STUDIES OF LSS PARAMETERIZATION, EVALUATED ON MATRIXCITY [31]. BEST RESULTS HIGHLIGHTED IN BOLD.

Method	Ablation	SSIM ↑	PSNR ↑	LPIPS ↓
Ablation on LSS Parameterization				
$\mathcal{N}(0, \sigma_{mc})$	w/o f_θ, w/o g_θ	0.586	16.225	0.480
$\mathcal{N}(0, \sigma_\theta)$	w/o f_θ, w/ g_θ	**0.609**	16.544	0.474
$\mathcal{N}(\mu_\theta, \sigma_{mc})$	w/ f_θ, w/o g_θ	0.594	16.599	0.478
$\mathcal{N}(\mu_\theta, \sigma_\theta)$	w/ f_θ, w/ g_θ	0.592	**16.693**	**0.470**

2) LSS Component Ablation: We ablate the two main components of the proposed LSS process, namely the CPF network f_θ and the variance network g_θ, with results presented in Tab. V. In the table, 3DGS-MCMC [5] is denoted by $\mathcal{N}(0, \sigma_{mc})$, where σ_{mc} is the hand-crafted variance. For the $\mathcal{N}(0, \sigma_\theta)$ (without CPF), the support input is replaced by a constant and the gate is conditioned on opacity and Gaussian state only. Replacing the hand-crafted variance with this learned stochastic proposal improves SSIM, PSNR, and LPIPS over the MCMC-variance row, indicating that adaptive stochasticity is beneficial under biased viewpoints. Introducing the CPF-conditioned mean term μ_θ while keeping the original variance, denoted by $\mathcal{N}(\mu_\theta, \sigma_{mc})$, improves PSNR over both variance-only rows, indicating that the CPF model provides observed-support guidance beyond pixel-wise supervision. The full LSS process, denoted by $\mathcal{N}(\mu_\theta, \sigma_\theta)$, achieves the best PSNR and LPIPS, while the learned-variance-only variant obtains the highest SSIM. This suggests that CPF drift improves photometric fidelity and artifact suppression, whereas stochastic adaptation better preserves local structural similarity.

TABLE VI
ABLATION STUDIES OF GEOMETRIC FIELD AND CPF MODEL, EVALUATED ON TANKS & TEMPLES [29] WITH COVERAGE GAP OF 60°.

Ablation	SSIM ↑	PSNR ↑	LPIPS ↓
Ablation on DBSCAN ϵ value			
$\epsilon = 0.1$	0.7058	19.1690	**0.2458**
$\epsilon = 0.25$	**0.7112**	**19.4428**	0.2461
$\epsilon = 0.5$	0.7058	19.1871	0.2485
$\epsilon = 0.75$	0.7048	18.9888	0.2527
$\epsilon = 1.0$	0.7004	18.9240	0.2550
Ablation on Network Architecture of CPF Model			
MLP	0.7070	19.2183	0.2463
SIREN-style	**0.7112**	**19.4228**	**0.2461**

3) CPF Construction Sensitivity: The effectiveness of the CPF model determines the quality of μ_θ. We therefore ablate two deciding factors of the CPF model, namely the probabilized geometric field and the CPF network architecture. Since the density and accuracy of the geometric field are determined by the DBSCAN parameter ϵ which controls the maximum neighborhood distance, we first ablate different ϵ values to assess the effect of geometric field quality on CPF performance. Results in Table VI show that performance peaks around $\epsilon = 0.25$ and gradually drops as ϵ increases, except for $\epsilon = 0.1$. This trend has two likely causes. First, when ϵ is below a certain threshold, DBSCAN fails to capture sufficient clusters, leading to too sparse or even empty geometric field that dampens CPF supervision. Second, larger ϵ values commonly lead to noisier clusters, which may introduce inaccurate guidance for CPF training. Based on these observations, we choose $\epsilon = 0.25$ for the best overall performance trade-off.

The second part of Table VI compares different network architectures for the CPF model. The SIREN [27]-style network achieves slightly better performance than a standard MLP, which suggests that CPF training is not highly sensitive to the network design. Given this advantage at a comparable model size, we adopt the SIREN-style network in our method.

TABLE VII
RESULTS OVER FIVE INDEPENDENT TRIALS ON TANKS & TEMPLES [29] WITH A 60° COVERAGE GAP.

Trial	SSIM ↑	PSNR ↑	LPIPS ↓
Trial 1	0.7052	19.3795	0.2476
Trial 2	0.7030	19.1766	0.2514
Trial 3	0.7070	19.3917	0.2486
Trial 4	0.7040	19.3068	0.2461
Trial 5	0.7075	19.3454	0.2474
Mean	0.7053	19.3200	0.2482
Variance	3.68×10^{-6}	0.0075	3.95×10^{-6}

4) LSS Stochastic Stability: To evaluate the sensitivity of our method to stochastic optimization, we repeatedly train GS-LSS with the same settings for five independent trials. Results in Table VII show only minor performance fluctuations across trials. The PSNR of each trial stays within approximately ± 0.15 dB of the mean, while the SSIM and LPIPS variances remain on the order of 10^{-6}. These results indicate that our method exhibits stable optimization behavior despite its stochastic nature.

TABLE VIII
RELATIVE TRAINING COST AND RENDERING QUALITY COMPARISON ON TANKS & TEMPLES [29] WITH A 60° COVERAGE GAP. TRAINING COST IS NORMALIZED BY VANILLA 3DGS.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	Time vs. 3DGS ↓
3DGS	18.651	0.695	0.269	1.00×
Scaffold-GS	18.658	0.692	0.269	1.29×
Hierarchical-GS	18.728	0.694	0.267	1.05×
3DGS-MCMC	18.973	0.702	0.257	1.24×
MH-3DGS	17.404	0.649	0.291	0.99×
PGSR	18.020	0.687	0.295	2.67×
MVGS	18.323	0.668	0.275	5.02×
GS-LSS (Ours)	**19.345**	**0.709**	**0.247**	2.77×

5) Efficiency Trade-off: Table VIII reports the quality-cost trade-off with training cost normalized by vanilla 3DGS [1]. Under the same 60° coverage-gap protocol, GS-LSS achieves the best PSNR, SSIM, and LPIPS at 2.77× the 3DGS training time. This cost is comparable to PGSR [2] (2.67×) and substantially below MVGS [21] (5.02×), while GS-LSS yields higher rendering quality than both. The extra time mainly comes from CPF training, CPF queries, and per-primitive LSS updates, which target weakly observed regions where biased training causes the largest degradation. These modules are used only during training. At inference, the CPF and stochastic heads are disabled, and rendering uses the standard optimized Gaussian field. We therefore view GS-LSS as a reasonable one-time training investment when held-out view fidelity and artifact suppression matter more than raw training speed.

V. LIMITATIONS

GS-LSS regularizes Gaussian primitives toward aggregated observed support, but it does not recover geometry that is completely unsupported by the input views. The CPF field is derived from filtered in-training geometry, so its reliability

depends on the quality of the stabilized Gaussian field and the effectiveness of outlier filtering. The method also increases training cost because of the CPF supervision and queries. Reducing this cost and improving CPF reliability under extremely sparse captures are important directions for future work.

VI. Conclusion

We present GS-LSS, an observed-support regularized learned stochastic optimization framework for Gaussian Splatting under biased viewpoints. Rather than relying solely on view-dependent photometric supervision, GS-LSS aligns Gaussian primitives with an aggregated observed-support field through a Learned Stochastic Sampling process. Its Continuous Probability Field estimates observed support from globally aggregated geometric priors and provides gradient-based drift with adaptive stochastic exploration. This scene-level guidance helps reduce perspective overfitting and encourages more globally consistent reconstruction. Experiments on three biased viewpoint scenarios show strong quantitative and qualitative results among the evaluated baselines, improving observed-support consistency at moderate training overhead. Future work will improve the efficiency and reliability of support estimation while maintaining high-quality rendering.

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